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ORIGINAL ARTICLE

The evolution of endemism within a major legume lineage of the *campos rupestres*

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• **Background and Aims** The *campos rupestres* (CR) are a grassy vegetation mosaic found on ancient mountains in Brazil. CR areas accumulate a high plant endemism that has been explained primarily by two contrasting processes, namely the Plio-Pleistocene climatic oscillations and long-term specialization and persistence in climatically stable areas. To investigate these alternative processes, we adopted the mountain-geobiodiversity (MG) hypothesis and studied the genus *Chamaecrista*, an emblematic plant lineage in CR areas. We hypothesized that species would have evolved towards environmental specialization upon colonizing CR areas because of selective pressures imposed by environmental heterogeneity. Moreover, species would have undergone faster diversification in CR areas during the Plio-Pleistocene, when climatic oscillations would have promoted speciation over extinction. Throughout, species would have originated primarily through geographical speciation in CR areas, where topographic complexity would have imposed physical barriers to plant dispersal.
• **Methods** We used geographical and phylogenetic data on the genus *Chamaecrista*, including CR-endemic and non-endemic species. We applied biogeographical reconstruction to infer when species became endemic to CR areas. We applied environmental niche modelling to estimate specialization, fitting evolutionary models to infer how specialization evolved. We fitted state-dependent models to infer the species diversification dynamic. We estimated the degree of geographical isolation during speciation.
• **Results** CR-endemic species evolved environmental niches twice as specialized as those of non-endemic species, probably under selective pressures. CR-endemic species experienced a faster diversification than non-endemic species at deeper evolutionary times, before the Plio-Pleistocene periods. CR-endemic species originated under greater geographical isolation than non-endemic species.
• **Conclusion** Colonization of CR areas accelerated species diversification, enhancing the pool of species later to become endemic. Environmental specialization has enforced plant endemism in CR areas by constraining the potential for persistence in habitats outside CR areas. Topographic complexity has enhanced plant endemism in CR areas by hindering plant dispersal and promoting geographical speciation.

Key words: *Chamaecrista*, climatic oscillation, diversification, environmental specialization, mountain, rocky grassland, topographic complexity.

INTRODUCTION

The *campos rupestres* (CR) are outstanding centres of plant endemism in the Neotropics (Alves and Kolbek, 1994; Colli-Silva *et al.*, 2019), whose evolutionary origins have long been debated (Fiaschi and Pirani, 2009). CR areas are mosaics of grassy vegetation and rocky outcrops that occur over ancient mountain regions (>900 m), primarily in Eastern and Central Brazil (Vasconcelos, 2011; Alves *et al.*, 2014; Fig. 1A–C). These mountain regions were formed during the Archaean and Palaeoproterozoic, and long-term erosion and episodic tectonics have created a complex topography (Saadi, 1995), including valleys, ridges, peaks and plateaus. Currently, CR areas are under subtropical and arid climate zones, where seasonal variation imposes dry periods (Alvares *et al.*, 2014). CR areas overall have shallow and oligotrophic soils (Benites *et al.*, 2003), but with varying degrees of nitrogen and phosphorus limitation

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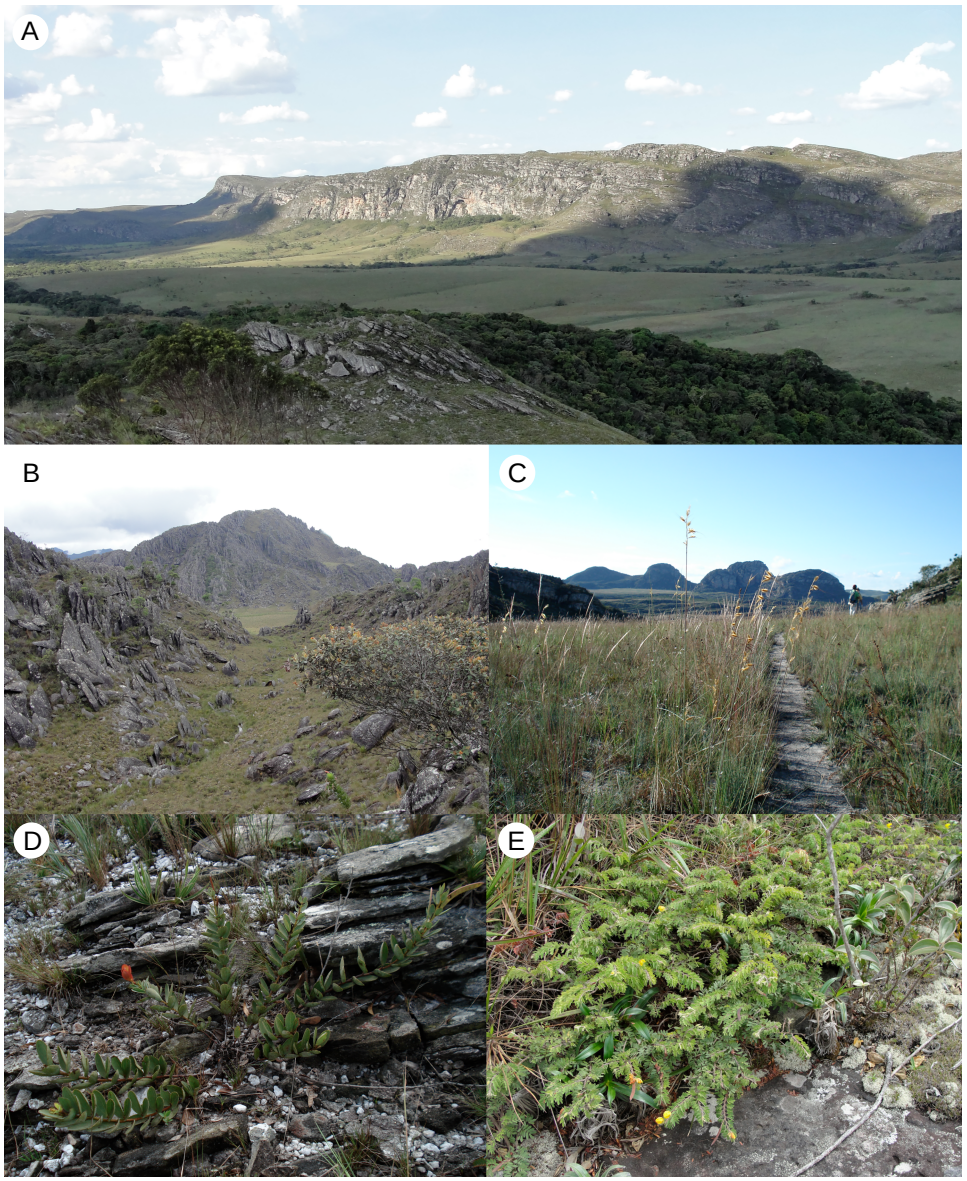

Fig. 1. Some areas of *campos rupestres* and typical *Chamaecrista* species in these areas. (A) Landscape showing the high-elevation areas of Serra do Cipó, Santana do Pirapama, Minas Gerais, Brazil. (B) At the summit of Serra das Almas, Rio de Contas, Bahia, Brazil. (C) Landscape showing the sandy soil on a trail in Diamantina, Minas Gerais, Brazil. (D) *Chamaecrista rossicorum* (H.S.Irwin & Barneby) Rando, sterile individual, Diamantina, Minas Gerais, Brazil. (E) *Chamaecrista arrojadoana* (Harms) Rando, in flower, Morro do Pai Inácio, Chapada Diamantina, Bahia, Brazil. Photographs: Juliana Gastaldello Rando.

(e.g. Oliveira *et al.*, 2015; Abrahão *et al.*, 2019). Fires can ignite in some CR areas during dry periods, yet without conditioning the maintenance of CR vegetation (Le Stradic *et al.*, 2018). These environmental conditions allow for a mosaic of habitat types within CR areas (Conceição *et al.*, 2016), including rock outcrops, clefts, bogs, grasslands and shrublands (e.g. Conceição and Pirani, 2005; Conceição *et al.*, 2007; Carmo and Jacobi, 2013). Reflecting their diversity of habitats, CR areas harbour ~5500 species of vascular plants (Flora e Funga do Brasil, 2025), of which 1750 (32 %) are endemic (Colli-Silva *et al.*, 2019). Many endemic plant species are restricted to a few or a single CR area, meaning that plant species composition differs drastically among neighbouring CR areas (e.g. Echternacht *et al.*, 2011). Hence, each CR area becomes a unique reservoir of plant endemism, raising questions about their evolutionary origin.

Plant endemism in CR areas has been explained primarily by evolutionary processes based on two alternative hypotheses: (1) the Pleistocene refuge (PR); and (2) old climatically buffered infertile landscapes (OCBIL) (Haffer, 1969; Hopper, 2009). Under the former hypothesis, CR-endemic plant species would derive primarily from diversification bursts during the Plio-Pleistocene periods (5.3–0.01 Mya), when climatic oscillations would have imposed rapid contraction–expansion dynamics over plant populations, favouring a fast species diversification, i.e. a 'species-pump' effect (Vasconcelos *et al.*, 2020). Supporting this first perspective, the origin or the diversification of multiple

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CR-endemic clades coincides with the Plio-Pleistocene periods (e.g. Antonelli *et al.*, 2010; Ribeiro *et al.*, 2014; Bochorny *et al.*, 2022), yet without necessarily conforming to a 'species-pump' effect. Under the second hypothesis, CR-endemic plant species would derive primarily from edaphic specialization coupled with short-range dispersal, which would be selectively advantageous in the heterogeneous landscapes of CR areas (Silveira *et al.*, 2016), assuming a buffered effect from climatic oscillations (Rapini *et al.*, 2021). Supporting this second perspective, plant communities have distinct functional traits among soil types in CR areas (e.g. Oliveira *et al.*, 2015; Brum *et al.*, 2017; Abrahão *et al.*, 2019), with a prevalence of autochoric dispersal modes (e.g. Arruda *et al.*, 2021), but the evolutionary origin of these functional traits remains largely untested. Thus, recent climatic oscillations and ancient edaphic specialization have been invoked to explain plant endemism in CR areas.

Albeit conflicting at the first glance, these alternative explanations are complementary processes under the mountain-geobiodiversity (MG) hypothesis (Mosbrugger *et al.*, 2018). Originally proposed for the Tibeto-Himalayan highlands, the MG hypothesis invokes three core processes to explain plant diversity in mountain regions (Muellner-Riehl, 2019). First, the environmental heterogeneity of mountains would impose distinct selective pressures on plant populations, continuously favouring the evolution of specialized plant species through geological time (Muellner-Riehl, 2019). Second, climatic oscillations would impose contraction–expansion dynamics over plant populations in mountain regions, promoting a 'species-pump' effect during specific time periods (Muellner-Riehl, 2019). Third, the topographic complexity of mountains would hinder dispersal and reduce the spatial connectivity among plant populations, continuously facilitating geographical speciation through geological time (Muellner-Riehl, 2019). Hence, the MG hypothesis combines the time-restricted effects of climatic oscillation invoked by the PR hypothesis and the long-term effects of environmental specialization proposed by the OCBIL hypothesis, while explicitly incorporating the long-term effects of topographic complexity. These evolutionary processes have been supported, in part, by a direct association between plant species richness and topographic variation in mountain regions worldwide (Muellner-Riehl *et al.*, 2019), yet without considering CR areas. Nonetheless, CR areas are marked by both palaeo- and neo-endemism (Barros-Souza and Borges, 2023), suggesting that ancient persistence and recent diversification might have promoted plant endemism in CR areas.

In the present study, we investigated whether the MG hypothesis could explain the evolution of plant endemism in CR areas, using the genus *Chamaecrista* (L.) Moench (Fabaceae) as a biological model. We hypothesized that the environmental heterogeneity within CR areas imposed selective pressures favouring environmental specialization among plant species. Furthermore, we hypothesized that Plio-Pleistocene climatic oscillations imposed a 'species-pump' effect on plant species in CR areas, while the topographic complexity of CR areas created dispersal barriers, thereby facilitating geographical speciation. Based on this hypothesis, we expected that: (1) CR-endemic species would have selectively evolved a greater degree of environmental specialization than non-endemic species; (2) CR-endemic species would have greater diversification rates than non-endemic species during the Plio-Pleistocene; and (3) CR-endemic species would have originated under greater geographical isolation than non-endemic species.

MATERIALS AND METHODS

We conducted all analytical procedures in the R environment v.4.5.0 (R Development Core Team, 2026). Throughout, we cited only the main functions and their respective packages. All data and scripts are available at: eknery/Chamaecrista_endemism_CR.

Focal study group

Our focal study group was *Chamaecrista* (L.) Moench (Caesalpinioideae, Fabaceae; Fig. 1D, E), a pantropical plant lineage with ~368 species, whose highest diversity (60 %) is concentrated in the Neotropics (Rando *et al.*, 2024). *Chamaecrista* species are woody plants with varying life forms, including annual and perennial life cycles, inhabiting rainforests and open-vegetation habitats (Rando *et al.*, 2024). Most *Chamaecrista* species depend on female vibrating bees for pollination (Gottsberger and Silberbauer-Gottsberger, 1988; Barbosa *et al.*, 2023), which can maintain long-distance gene flow (e.g. Fenster *et al.*, 2003). Moreover, *Chamaecrista* species produce autocoric pods (Rando *et al.*, 2024), which can disperse seeds by 2–3 m (e.g. Fenster, 1991). All analysed *Chamaecrista* species so far can host nitrogen-fixing bacteria in their root systems (Casaes *et al.*, 2024), some of which have a strict dependence on these bacteria (e.g. Souza *et al.*, 2024, 2025).

In the Neotropics, the taxonomic diversity of *Chamaecrista* has been the subject of extensive systematic studies (de Souza Conceição *et al.*, 2009; Rando *et al.*, 2016; de Souza *et al.*, 2019; Cota *et al.*, 2025). The Neotropical *Chamaecrista* species had their evolutionary relationships elucidated in a recent phylogenetic analysis by Rando *et al.* (2026; Supplementary Data Fig. S1). This phylogenetic hypothesis includes 231 (63 %) of the 368 *Chamaecrista* species, representing: 167 (89 %) of the 186 species endemic to Neotropical open-vegetation domains; 17 (62 %) of the 27 species endemic to Neotropical rainforest domains; and 47 species with widespread distribution (Rando *et al.*, 2026). We randomly sampled 100 phylogenetic hypotheses from the posterior distribution inferred by Rando *et al.* (2026) to run our analyses, using their maximum clade credibility (MCC) phylogenetic hypothesis for graphical display.

Environmental variables

To represent the environmental gradients in our study area, we used global raster layers from data repositories. We selected four raster layers: the s.d. of annual temperatures (temperature seasonality, in degrees Celsius); the total precipitation during the warmest quarter (precipitation, in millimetres); the mean nitrogen per soil mass from 0 to 30 cm depth (soil nitrogen, in grams per kilogram); and

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the number of fires from 2015 to 2024 (fire frequency, expressed as fires per decade). We retrieved precipitation and temperature rasters from WorldClim 2.0 (Fick and Hijmans, 2017), soil nitrogen from Wosis (Poggio *et al.*, 2021), and fire frequency from MODIS C.61 (Giglio *et al.*, 2016). We standardized all rasters to the WGS84 coordinate reference system and to the 30″ resolution, setting ~1 km^2 grid cells (sites). These raster layers represented environmental gradients with distinct conditions between CR and surrounding areas (Silveira *et al.*, 2016). Moreover, some of these environmental gradients have previously been inferred to affect the evolution of plant life forms in *Chamaecrista* (Prochazka *et al.*, 2024), thus representing key niche axes for our focal study group. Finally, we retrieved the elevation raster layer from WorldClim 2.0 to assess species endemism status in CR areas.

Gathering geographical records and defining states

To allow species geographical classification and environmental niche modelling, we assembled a dataset of georeferenced records based on validated herbarium vouchers. We searched records for all species represented in the phylogenetic hypothesis of *Chamaecrista* ($n = 231$; Rando *et al.*, 2026), considering their accepted names, subordinated taxa and synonyms. We retrieved records from the Global Biodiversity Inventory Facility (GBIF), by using the R package *plantR* (de Lima *et al.*, 2023). We selected only records linked to herbarium vouchers that have been identified by an acknowledged taxonomist of *Chamaecrista*. We treated all records with standard procedures by using the R package *CoordinateCleaner* (Zizka *et al.*, 2019). We thinned the records of each species by excluding points in a same site (grid cell) to mitigate spatial sampling bias over niche modelling. We also retrieved records from the curated dataset of Prochazka *et al.* (2024).

To classify species into geographical states with respect to CR areas, we considered two criteria based on taxonomic expertise and elevation range. First, we assessed whether *Chamaecrista* species have been recorded in CR areas and other vegetation types based on their most recent taxonomic monograph (Rando *et al.*, 2025). Second, we evaluated whether *Chamaecrista* species occupied median elevations ≥900 m, matching the elevation range of CR areas (Vasconcelos, 2011; Alves *et al.*, 2014). We measured the median elevation of species by projecting their curated geographical records on the elevation raster from WorldClim 2.0 (Fick and Hijmans, 2017), under the coordinate system and resolution previously elaborated. Hence, we classified species as 'CR-endemic' when recorded only in CR areas and matching the elevation range of CRs; 'CR and other areas' when recorded only in CR areas but below the elevation range of CRs or when recorded in CR and other areas and matching the elevation range of CRs; and 'outside CR' when never recorded in CR areas regardless of elevation range.

Biogeographical reconstruction

To infer when ancestral species colonized CR areas and became endemic, we applied a biogeographical reconstruction to our sample of phylogenetic hypotheses ($n = 100$), by using the GeoHiSSE algorithm (Caetano *et al.*, 2018). Under the GeoHiSSE algorithm, species can disperse between two areas at certain rates (d) and evolve geographical states, which can then affect the net diversification (τ) and relative extinction (ϵ) rates (Caetano *et al.*, 2018). Moreover, hidden states (i.e. unknown factors) can be added to detect heterogeneous diversification rates unrelated with geographical states (Caetano *et al.*, 2018). By incorporating diversification, the GeoHiSSE algorithm improves biogeographical reconstructions, especially in highly asymmetrical extinction scenarios (Sanmartín and Meseguer, 2016).

We considered two geographical areas, 'CR' and 'other areas', consequently allowing three possible geographical states: 'CR-endemic', 'CR and other areas' and 'outside CR'. We compared two GeoHiSSE models with equal complexity (number of parameters): (1) GeoHiSSE1, with variable dispersal rates among geographical states, with fixed net diversification and relative extinction rates among geographical states, but with variable rates among hidden states; and (2) GeoHiSSE2, with variable dispersal rates among geographical states, variable net diversification and relative extinction rates among geographical states, without hidden states. We disregarded models with homogeneous diversification rates because they are often unrealistic and become trivial null models (Beaulieu and O'Meara 2016). The two models did not allow direct changes between 'CR-endemic' and 'outside CR' states, thus representing a gradual colonization scenario. We also incorporated the relative species diversity (62 %) into parameter estimation. We compared the goodness of fit by the relative weights of the corrected Akaike information criterion (AICcw). We chose the GeoHiSSE model with the most frequent best fit for our sample of phylogenetic hypotheses.

Environmental niche modelling

To estimate the environmental specialization of species, we applied the hypervolume modelling method in the R package *hypervolume* (Blonder *et al.*, 2014), following the protocol of Nery *et al.* (2023). Specialization designates a limited breadth over a single or multiple niche axes, translating into a reduced n-dimensional niche (Futuyma and Moreno, 1988). The hypervolume method models the n-dimensional niche of species over environmental gradients by applying kernel density estimation (KDE) on occupied environmental values of species (Blonder *et al.*, 2014). The resulting model has a multidimensional volume, the hypervolume, whose size inversely quantifies the niche breadth over multiple environmental axes (Blonder *et al.*, 2014). We retrieved the occupied environmental values of species by projecting the georeferenced records onto the four raster layers. We also defined unoccupied environmental values based on the background-group strategy (Ponder *et al.*, 2001). Under this strategy, a species is considered absent in sites that do not have their records but that have records of other related species, i.e. the background group (Ponder *et al.*, 2001).

Prior to modelling, we evaluated the predictive performance of hypervolume models by a k-fold validation. We considered sites with species records as 'presences' and sites

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from the respective background group as 'absences'. We split presences into three sets ($k = 3$) for common species (n records ≥ 12), using two sets for training and one set for testing in each round. We applied a leave-one-out procedure ($k = n$ records) for rare species ($8 \leq n$ records < 12). Species with too few records (n records < 8) were excluded from niche modelling. We built hypervolume models by using a Gaussian estimator with a normal bandwidth (Blonder *et al.*, 2018). We considered distinct probability thresholds to delimit the hypervolume (0.05, 0.25, 0.5, 0.75 and 0.95) for each fold. We calculated the predictive performance of hypervolume models by the true skill statistics (TSS) (Allouche *et al.*, 2006). We chose the model settings that maximized TSS values for each species.

After validation, we built hypervolume models for each species by using all their occupied environmental values and considering their best model settings. We disregarded species whose hypervolume models had very low performance (TSS < 0.1) during validation. We used the hypervolume size as a proxy for the environmental niche breadth, where smaller hypervolume sizes indicated greater specialization (Blonder, 2018). Nonetheless, occurrence-based niche models can confound niche breadth and geographical range under scenarios of spatial autocorrelation (Cardillo *et al.*, 2019), in which niche breadth increases only with the addition of spatially distant areas. Hence, we evaluated whether hypervolume dimensions were inversely correlated with spatial correlation, by using built-in linear models. We estimated spatial autocorrelation with the local Moran's I among neighbouring sites (3×3 window), by using the R package *terra* (Hijmans, 2020). We did not find a significant correlation between hypervolume dimensions and Moran's I (Supplementary Data Fig. S2), hence we considered the hypervolume size a suitable measure of environmental specialization.

Inferring the evolution of environmental niche breadth

To infer how the environmental niche breadth evolved in CR-endemic and non-endemic species (first prediction), we fitted evolutionary models to hypervolume size across our sample of phylogenetic hypotheses ($n = 100$). We considered the evolutionary models for continuous phenotype based on Brownian motion (BM) and Ornstein–Uhlenbeck (OU) process, as implemented in the R package *Ouwie* (Beaulieu *et al.*, 2012). BM models describe phenotypic evolution as a stochastic process at a certain rate (σ^2) (Felsenstein, 1988). OU models also describe phenotypic evolution as a stochastic process at a certain rate (σ^2), while additionally considering a dragging process towards an optimal value (θ) under a certain strength (α) (Hansen, 1997). In short, BM models represent neutral evolution or inconsistent selective pressures, whereas OU models represent consistent selective pressures or some sort of constraint (Butler and King, 2004).

We considered eight alternative models to investigate how hypervolume size evolved with respect to geographical states of ancestral and extant species. We considered two BM and five OU models whose parameters could be either fixed or variable among geographical states (Supplementary Data Table S1). We also considered the white noise (WN) model to represent a non-evolutionary scenario for hypervolume variation. We used the best-fitting GeoHiSSE model to assign ancestral species to geographical states. Prior to model fitting, we cubic root-transformed hypervolume size to attain linearity and residuals with a normal distribution. We compared the goodness of fit by the corrected Akaike information criterion (AICc) (Sugiura, 1978), assuming a minimum difference of two scores to favour a more complex model over a simpler one. We chose the evolutionary model with the most frequent best fit for our sample of phylogenetic hypotheses.

It is noteworthy that we interpreted these evolutionary models as proxies for species-level regimes that would derive from population-level processes (Hansen and Martins, 1996). If populations face stronger divergent selection inside CR areas than outside them, they are more likely to originate specialized species inside CR areas. This first scenario predicts a selective regimen favouring increased specialization for species within CR areas. Conversely, if populations face similar strengths of divergent selection inside and outside CR areas, they are equally likely to originate specialized species regardless of geographical area. This second scenario predicts selective regimes favouring a similar degree of specialization for species inside and outside CR areas.

Inferring evolutionary correlates of range size

To infer whether geographical range size evolved with environmental niche breadth, we fitted a phylogenetic generalized least square (PGLS) model. The PGLS models allow evaluation of whether a response trait had evolved jointly with other predictor traits within a given clade, i.e. evolutionary correlation (Harmon, 2019). The response trait was species geographical range size, calculated as the convex hull area delimited by the geographical records (Darroch and Saupe, 2018), and the predictor trait was species environmental niche breadth, represented by the hypervolume size from niche models (Blonder *et al.*, 2014). Nonetheless, geographical range size tends to increase with species evolutionary age (Alzate *et al.*, 2025), reflecting an overall geographical expansion over generations. Hence, we also included species evolutionary age as a predictor trait, represented by the node height in the MCC phylogenetic hypothesis. We log-transformed the convex hull area to attain linearity and residuals with a normal distribution. We transformed predictor trait values to z-scores in order to directly compare their relative effects by the standardized slope coefficients.

Inferring diversification dynamics

To infer how fast CR-endemic and non-endemic species diversified (second prediction), we used two types of hidden state speciation and extinction (HiSSE) algorithms. We used the GeoHiSSE algorithm to infer whether diversification rates differed among ancestral species in distinct geographical states, investigating the diversification dynamics at deep evolutionary times (Caetano *et al.*, 2018). Conversely, we used the Missing State Speciation Extinction (MiSSE)

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algorithm to infer whether diversification rates differed among extant species, investigating the diversification dynamics at recent evolutionary times (Vasconcelos *et al.*, 2022). Throughout, we only considered estimates of the net diversification rate, because extinction rates are poorly estimated from extant-only phylogenetic hypotheses (Rabosky, 2010).

First, we fitted two alternative GeoHiSSE models to our sample of phylogenetic hypotheses ($n = 100$), as already elaborated (see '*Biogeographic reconstruction*'). The GeoHiSSE1 had fixed rates of net diversification and relative extinction among geographical states, but not among hidden states, representing a scenario in which CR areas did not affect the diversification dynamics; the GeoHiSSE2 model had variable rates of net diversification and relative extinction among geographical states, representing a scenario in which CR areas affected the diversification dynamics. We retrieved estimates of net diversification from the best-fitting GeoHiSSE model and compared them among CR-endemic and non-endemic species.

Second, we fitted eight alternative MiSSE models only to the MCC phylogenetic hypothesis, for computational reasons. Each MiSSE model considered a different number of hidden states (ranging from zero to eight) that had variable rates of net diversification and relative extinction, representing a range from strict homogeneous to highly heterogeneous diversification dynamics. We compared the goodness of fit by the AICcw. We retrieved estimates of net diversification rates from the best-fitting MiSSE model and weighted them by the relative probability of each hidden state.

Inferring the time and mode of speciation

To infer when and how CR-endemic and non-endemic species underwent speciation (third prediction), we calculated the mean node height and the range overlap (RO) intercept from our sample of phylogenetic hypotheses ($n = 100$). First, we calculated the mean node height for each extant species to assess whether their speciation occurred during the Plio-Pleistocene. Second, we calculated the RO intercept among extant species to assess whether their speciation occurred by geographical isolation. This metric is the intercept estimate of the linear regression between the geographical overlap of sister lineages and their respective divergence time (Skeels and Cardillo, 2019). This metric can distinguish past speciation scenarios based on current geographical distributions, with a low interference from other factors affecting geographical distribution (Skeels and Cardillo, 2019). In short, smaller RO intercept estimates indicate greater geographical isolation during speciation. We measured the RO between every pair of extant sister lineages based on the intersection of their convex hull areas, using the R package *sf* (Pebesma, 2018). Finally, we estimated the RO intercept from linear regressions and contrasted their values between CR-endemic and non-endemic species.

RESULTS

We reconstructed the biogeographical history of 231 *Chamaecrista* species based on the best-fitting GeoHiSSE model (Fig. 2A), considering three geographical states: CR-endemic ($n = 56$); CR and other areas ($n = 74$); and outside CR areas ($n = 101$). The last common ancestor of *Chamaecrista* was likely to be outside CR areas ~42 Mya. Three ancestral lineages from outside areas independently dispersed to CR areas between 29 and 9 Mya, giving rise to three ancestral lineages in CR and other areas. Later, 15 ancestral lineages in CR and other areas underwent local extinction in CR areas and became restricted to outside areas from 15 Mya onwards. Multiple ancestral lineages in CR and other areas were extinct in outside areas and became endemic to CR areas from 12 Mya onwards.

We assembled a curated dataset of 9061 geographical records for 231 *Chamaecrista* species, including their subordinated taxa and respective synonyms (Fig. 2B). This curated dataset included only unique records per site for each species, thereby minimizing sampling bias. The number of records per species was similar among CR-endemic species (median ± interquartile range: 34 ± 42), species in CR and other areas (33 ± 49) and species outside CR areas (33 ± 60) (Supplementary Data Table S2).

We found that *Chamaecrista* species occupy higher elevations towards CR areas (Fig. 2C), based on our curated dataset with unique records per site. The elevation range of CR-endemic species (median ± interquartile range: 1176 ± 85 m) was 30–100 % higher than those of species in CR and other areas (903 ± 222 m) and species outside CR areas (590 ± 342 m), respectively.

We built reliable hypervolume models for 107 *Chamaecrista* species, including CR-endemic species ($n = 19$), species in CR and other areas ($n = 39$) and species outside CR ($n = 49$) (Supplementary Data Table S2). The hypervolume size of CR-endemic species (mean ± s.d.: 2.20 ± 2.76) was two times smaller than those of species in CR and other areas (5.33 ± 5.40) and species outside CR areas (5.79 ± 6.15), indicating greater environmental specialization among CR-endemic species (Fig. 2D).

The evolution of environmental niche breadth

The OU model was the best fit for the hypervolume size of *Chamaecrista* species in 36 % of phylogenetic hypotheses, followed by the OUMVA model in 26 %. Despite its good fit, we disregarded the OUMVA model because of its poor parameter estimates, a recurrent issue for this complex model (Cressler *et al.*, 2015). Under the OU model, phenotypic evolution follows a stochastic process with a fixed rate (σ^2) and a dragging process towards distinct optimal values (θ) under a fixed strength (α), supporting a selection-driven evolutionary scenario (Beaulieu *et al.*, 2012). The θ estimates of hypervolume size for CR-endemic species (mean ± s.d.: 1.41 ± 0.11) were two times smaller than those of species in CR and other areas (3.76 ± 0.22) and outside CR areas (3.93 ± 0.05) (Fig. 3A). The σ^2 (6.24 ± 7.65) and α (17.7 ± 21.7) estimates did not vary among geographical states.

Evolutionary correlates of range size

The convex hull area was weakly correlated ($r = 0.23$) with the hypervolume size of *Chamaecrista* species

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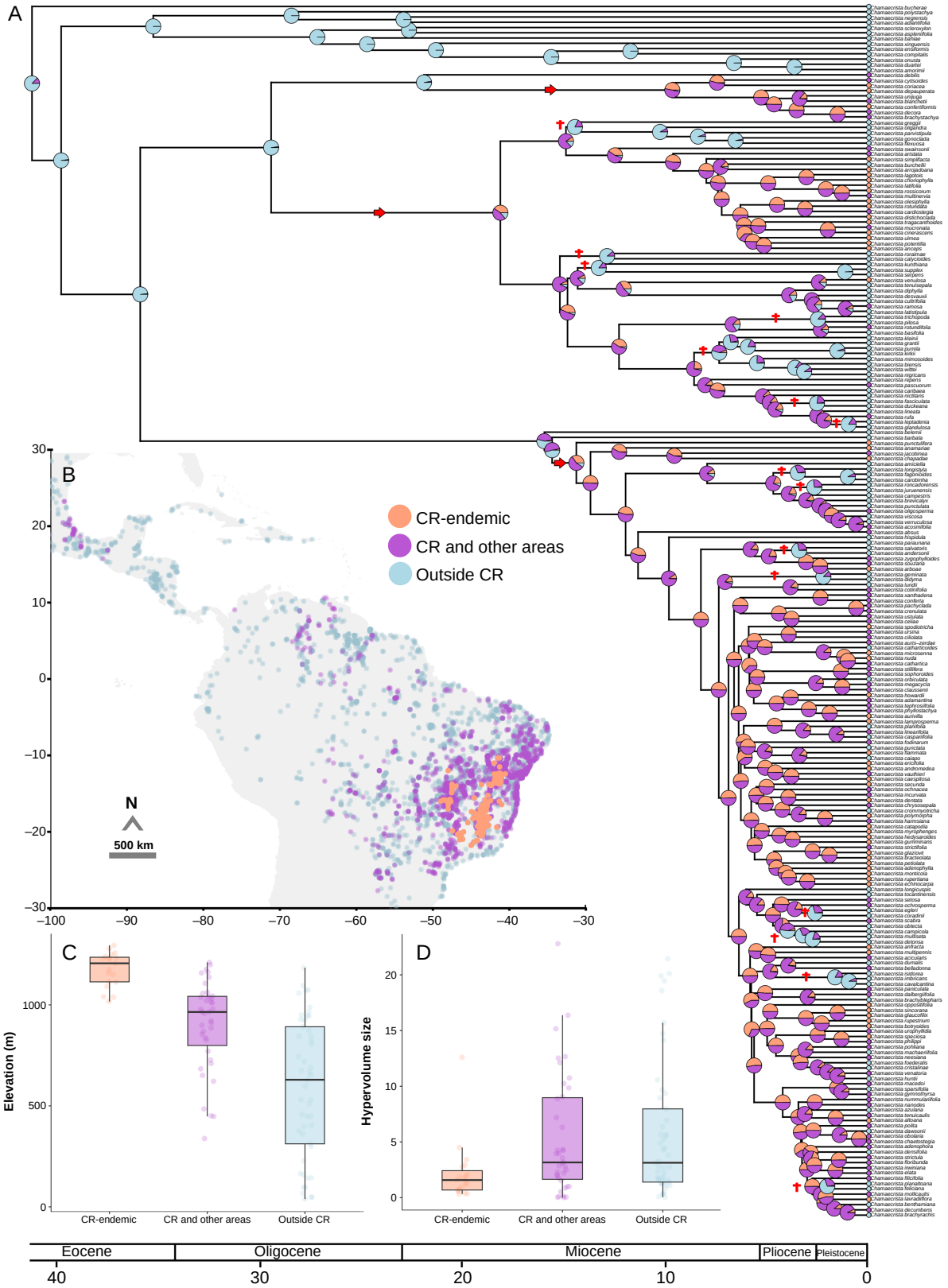

FIG. 2. The biogeographical reconstruction and geographical–environmental patterns of extant *Chamaecrista* species. (A) The biogeographical reconstruction on the MCC phylogenetic hypothesis, following the best-fitting GeoHiSSE model. Pie charts at nodes indicate the relative probability of each geographical state. Red arrows indicate ancestral dispersals to CR areas, while red crosses indicate ancestral extinctions in CR areas. The scale at the bottom indicates time before present, in million years. (B) The curated geographical records of species in the Neotropics. (C) The median elevation occupied by species in each geographical state. (D) The hypervolume size, a proxy for species' environmental niche breadth, in each geographical state.

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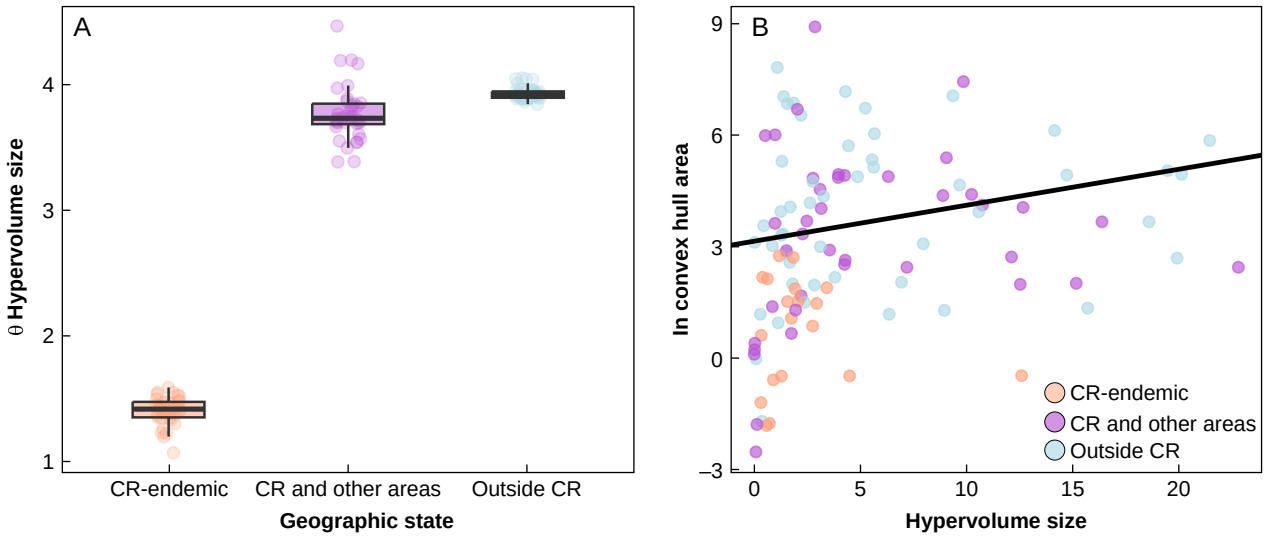

FIG. 3. The evolutionary optimum of hypervolume size, a proxy for species environmental niche breadth, and its evolutionary correlation with the convex hull area, a proxy for species geographical range, among *Chamaecrista* species in each geographical state. (A) θ estimates of hypervolume size for each geographical state based on our sample of phylogenetic hypotheses ($n = 100$), following the best-fitting Ornstein–Uhlenbeck model. (B) The direct evolutionary correlation between hypervolume size and log-transformed convex hull area, based on a PGLS model fitted to the MCC phylogenetic hypothesis.

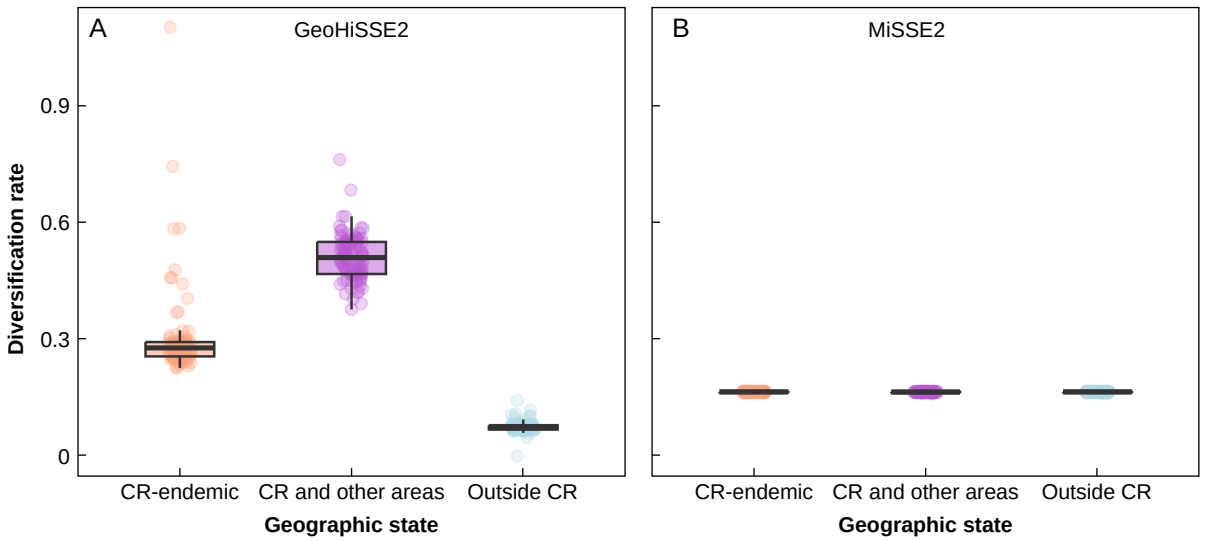

FIG. 4. Estimates of net diversification rates for *Chamaecrista* species in each geographical state. (A) Estimates of net diversification rates from our sample of phylogenetic hypotheses ($n = 100$), following the best-fitting GeoHiSSE model. (B) Estimates of net diversification rates from the MCC phylogenetic hypothesis, following the best-fitting MiSSE model.

(slope = 0.535, t-value = 2.653, $P < 0.01$; Fig. 3B), but uncorrelated with phylogenetic node height (slope = 0.183, t-value = 0.672, $P = 0.502$). This pattern indicates an evolutionary correlation between geographical range size and environmental niche breadth, without an effect from species evolutionary age.

Diversification dynamics

The GeoHiSSE2 model was the best fit for all phylogenetic hypotheses. Under this model, the net diversification and relative extinction rates varied among geographical states, supporting diversification shifts with transitions among geographical areas. The net diversification rates of CR-endemic species (mean ± s.d.: 0.313 ± 0.118) were four times greater than those of species outside CR areas (0.076 ± 0.015), but smaller than those of species in CR and other areas (0.526 ± 0.062) (Fig. 4A).

The MiSSE2 model was the best fit to the MCC phylogenetic hypothesis. Under this model, the net diversification and relative extinction rates varied with two hidden states, supporting heterogeneous diversification dynamics. Extant CR-endemic and non-endemic species had similarly low net diversification rates (mean ± s.d.: 0.1645 ± 0.0004; Fig. 4B), whereas some ancestral species had extremely low net diversification rates (0.004 ± 0.001) between 7.5 and 2.5 Mya (Supplementary Data Fig. S3).

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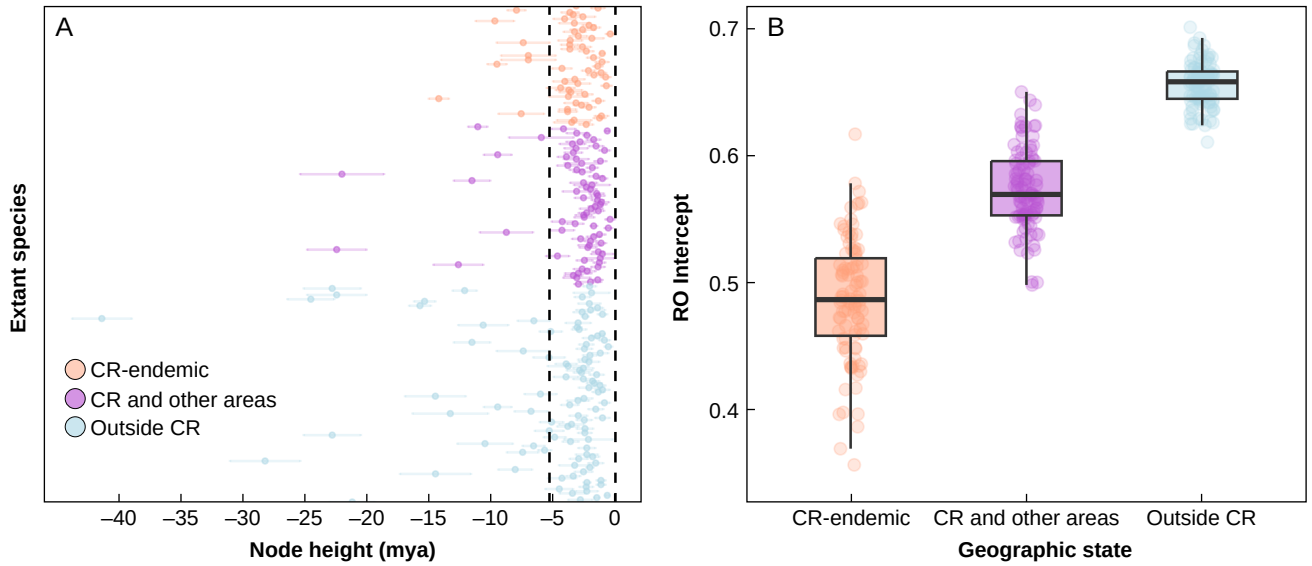

FIG. 5. The time and mode of speciation of extant *Chamaecrista* species in each geographical state. (A) The mean (dots) and s.d. (horizontal bars) of node heights for each extant species based on our sample of phylogenetic hypotheses ($n = 100$). The vertical dashed lines encompass the Plio-Pleistocene periods. (B) The RO intercept estimates, a proxy for sympatric speciation, for each geographical state based on our sample of phylogenetic hypotheses ($n = 100$).

Time and mode of speciation

Phylogenetic nodes of extant *Chamaecrista* species had an overall height of 4.3 Mya, indicating their speciation during the Plio-Pleistocene periods (Fig. 5A). The RO-intercept estimates of CR-endemic species (mean ± s.d.: 0.48 ± 0.04) were 15–30 % smaller than those of species in CR and other areas (0.57 ± 0.03) and species outside CR areas (0.65 ± 0.01), indicating speciation with greater geographical isolation in CR areas (Fig. 5B).

DISCUSSION

Our results provided partial support to the MG hypothesis in the CR, confirming environmental specialization and geographical isolation as major drivers of plant endemism in CR areas. Supporting our first prediction, we found that CR-endemic species evolved environmental niches twice as specialized as their non-endemic congeners, underscoring the role of selective pressures within CR areas. This pronounced environmental specialization was evolutionarily correlated with smaller geographical ranges, a pattern consistent with adaptation to a mosaic of habitat types. Countering our second prediction, CR-endemic species did not experience faster diversification during the Plio-Pleistocene periods, ruling out a 'species-pump' effect in CR areas during recent climatic oscillations. Nonetheless, CR-endemic species experienced faster diversification at deeper evolutionary times, probably upon colonizing CR areas. Supporting our third prediction, CR-endemic species originated under greater geographical isolation than their non-endemic congeners, confirming topographic complexity as a major promoter of geographical speciation in CR areas. Ultimately, our findings suggest that ancestral diversification bursts, environmental specialization and topographic complexity have contributed jointly to the evolution of CR-endemic plant species, broadening our current theoretical perspective on how plant endemism has evolved in CR areas.

Environmental specialization enforces plant endemism in CR areas

Supporting our first prediction, CR-endemic *Chamaecrista* species evolved environmental niches two times more specialized than those of non-endemic species. This pattern indicates that CR areas have imposed selective pressures favouring greater environmental specialization among plant species than lowland areas, as proposed by the MG hypothesis (Mosbrugger *et al.*, 2018; Muellner-Riehl, 2019). Such an environmental specialization is likely to have evolved in response to environmental heterogeneity (Brown and Pavlovic, 1992), which is presumably greater in CR areas than in nearby lowland areas (Silveira *et al.*, 2016; Rapini *et al.*, 2021). Each CR area is a mosaic of distinct habitats, including rocky outcrops, bogs, grasslands and shrublands (Conceição *et al.*, 2016), potentially selecting for different sets of physiological tolerances and requirements. For instance, within a CR area of the *Serra do Cipó* mountain range, plant species growing in white-sand grasslands face greater soil nitrogen limitations than those in rocky outcrops (e.g. Abrahão *et al.*, 2019). Under this scenario, plant species are more likely to persist within CR areas by specializing in habitat types, thereby avoiding the costs of coping with contrasting environmental conditions (Ravigné *et al.*, 2009). Probably reflecting this process, CR-endemic *Chamaecrista* species have evolved distinct traits to cope with contrasting environmental conditions in CR areas. For instance, *Chamaecrista latifolia* (Benth.) Rando evolved a geophyte form that can resprout after fires, whereas *Chamaecrista venulosa* (Benth.) H.S.Irwin & Barneby evolved a chamaephyte form tolerant to low soil nitrogen, but sensitive to fires (Prochazka *et al.*, 2024). Additionally, CR-endemic *Chamaecrista* species have evolved distinct

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degrees of phenotypic integration with nitrogen-fixing bacteria (Casaes *et al.*, 2024), probably in response to the varying degrees of soil nitrogen limitations within CR areas. Thus, CR-endemic *Chamaecrista* species are likely to have evolved towards environmental specialization owing to the environmental heterogeneity within CR areas.

Environmental specialization evolved with smaller geographical ranges for *Chamaecrista* species. Specialized species are expected to find fewer suitable habitats when dispersing across landscapes, leading to more restricted geographical ranges (Brown, 1984). Additionally, specialized species are less likely to adapt to novel environments owing to evolutionary constraints imposed by past selective optimizations (Poisot *et al.*, 2011). Supporting this theoretical expectation, plant species with specialized climatic niches also have smaller geographical ranges (Sheth *et al.*, 2020). Nonetheless, other intrinsic and extrinsic factors can affect the geographical range of plant species, such as evolutionary age, seed dispersal mode and geographical barriers (Sheth *et al.*, 2020). Geographical range size was uncorrelated with divergence time for *Chamaecrista* species, indicating that evolutionary age has not constrained geographical range in this plant group. Furthermore, both CR-endemic and non-endemic *Chamaecrista* species share similar autochoric dispersal modes (Rando *et al.*, 2024), indicating that seed dispersal mode could not account for the smaller geographical ranges of this plant group in CR areas. Notably, CR-endemic *Chamaecrista* species exhibited smaller geographical ranges than non-endemic relatives with similar niche breadths, indicating that factors other than environmental specialization have constrained the geographical range of endemic species in CR areas (see '*Topographic complexity enhances plant endemism in CR areas*' below). Therefore, environmental specialization and topographic complexity have probably constrained the geographical range of CR-endemic *Chamaecrista* species.

Edaphic conditions constrain the geographical range of multiple CR-endemic plant species (Rapini *et al.*, 2021), indicating that environmental specialization is a major promoter of plant endemism in CR areas. Nonetheless, plant species (and lineage) turnover between CR areas and neighbouring domains is generally low (e.g. Neves *et al.*, 2018; Barros-Souza *et al.*, 2026), which has been interpreted as evidence countering environmental specialization in CR areas. However, these patterns derive from analyses that did not account for plant endemism status (e.g. Neves *et al.*, 2018; Barros-Souza *et al.*, 2026), hence they primarily reflect non-endemic plant species that constitute the majority (68 %) of CR flora (Colli-Silva *et al.*, 2019). Moreover, the current similarity in species composition does not reflect the ancestral dispersal directionality (Alzate and Hagen 2022), meaning that floristic affinities between CR and other domains can result from an asymmetric accumulation of plant lineages that dispersed to CR areas rather than in the opposite direction. Hence, environmental specialization would drive the evolution of plant endemism in CR areas, as assumed for other mountain regions worldwide (Rahbek *et al.*, 2019). For instance, in a specific clade of *Miconia* Ruiz & Pav. (Melastomataceae), species evolved greater environmental specialization upon colonizing the montane areas of the Atlantic Forest (Nery *et al.*, 2023), in addition to novel floral traits more suitable for this mountain region (Nery *et al.*, 2024), possibly compromising their potential to persist in nearby lowland areas. Likewise, environmental specialization has also been invoked to explain plant endemism in the Andes (Young *et al.*, 2002), the most endemic-rich mountain region (Rahbek *et al.*, 2019). Thus, we propose that environmental specialization has hindered CR-endemic *Chamaecrista* species from persisting and adapting in lowland habitats, consequently enforcing plant endemism in CR areas.

Fast diversification precedes plant endemism in CR areas

Countering our second prediction, CR-endemic *Chamaecrista* species diversified faster than non-endemic relatives at deeper evolutionary times (GeoHiSSE2 model), but not at recent evolutionary times (MiSSE2 model). This pattern indicates that Plio-Pleistocene climatic oscillations have not imposed a 'species-pump' effect on plant species in CR areas, deviating from the MG hypothesis (Mosbrugger *et al.*, 2018; Muellner-Riehl, 2019). The Plio-Pleistocene climatic oscillations have previously been inferred to accelerate the speciation rates of CR-endemic *Chamaecrista* species (Vasconcelos *et al.*, 2020). However, speciation rates alone cannot reflect the dynamic of species accumulation, which is given by speciation and extinction together, i.e. net diversification (Wiens, 2011). Hence, this previous pattern only partly represented the effect of Plio-Pleistocene climatic oscillations on CR-endemic *Chamaecrista* species. When considering other plant groups, the Plio-Pleistocene climatic oscillations had disparate effects on species diversification in CR areas. For instance, *Vellozia* Vand. (Velloziaceae) experienced a faster diversification upon colonizing CR areas during the early Miocene, without further diversification shifts during the Plio-Pleistocene (Alcantara *et al.*, 2018; Magri *et al.*, 2025). In contrast, *Cambessedesia* DC. (Melastomataceae) originated in CR areas during the early Miocene and experienced a faster diversification in the same areas during the early Pliocene (Bochorny *et al.*, 2022). These inconsistent temporal patterns can also indicate that Plio-Pleistocene climatic oscillations might have a weak effect on plant species diversification in CR areas. Supporting this alternative perspective, high-elevation areas in the Cerrado, including some CR areas, have been inferred to be more climatically stable than lowland ones (de Deus Vidal *et al.*, 2019). Consequently, woody plant species at higher elevations had maintained similar geographical distributions between the Last Interglacial and the Last Glacial Maximum, whereas woody plant species at lower elevations did not (Silva *et al.*, 2024). Thus, ancestral *Chamaecrista* species would have undergone mild contraction–expansion dynamics in CR areas during the Plio-Pleistocene, precluding a 'species-pump' effect.

Although countering a 'species-pump' effect, our findings are still consistent with the bottleneck effects on CR plant populations during the Pleistocene (Dantas-Queiroz *et al.*, 2023). Bottleneck effects are fast and drastic reductions of effective population size (Maruyama and Fuerst, 1985), which have been assumed to underlie species diversification in CR areas (Dantas-Queiroz *et al.*, 2023). First, bottleneck effects can promote geographical isolation among populations, which would facilitate geographical speciation (Yamaguchi *et al.*, 2026). Still, CR plant populations might have undergone

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mild size reductions (e.g. Fiorini *et al.*, 2019), precluding the long-term geographical isolation necessary for speciation (~2 Myr; Hedges *et al.*, 2015). Moreover, bottleneck effects can also compromise the persistence of populations, promoting extinction over speciation (Yamaguchi *et al.*, 2026), ultimately reducing species diversification. Second, bottleneck effects can erode the genetic potential for local adaptation (Leimu and Fischer, 2008), potentially hindering the evolution of environmental specialization (Ravigné *et al.*, 2009). Yet, bottleneck effects might have occurred only in certain CR plant populations, rather than affecting the full geographical range of CR plant species (e.g. Barres *et al.*, 2019), thus preserving the genetic potential for local adaptation. Additionally, when assuming a general bottleneck effect on most species, populations could experience relaxed competition and further expand into novel habitats, potentially undergoing local adaptation (Yamaguchi *et al.*, 2026). Hence, the Plio-Pleistocene climatic oscillations promoted bottleneck effects on CR plant populations, but without necessarily increasing plant species diversification or hindering environmental specialization.

Nonetheless, CR-endemic *Chamaecrista* species and their relatives in CR and other areas diversified faster than species outside CR areas at deeper evolutionary times. This pattern suggests that ancestral *Chamaecrista* species underwent a faster diversification upon colonizing CR areas between the early Oligocene and the middle Miocene. The colonization of mountains presumably promotes species diversification by facilitating both geographical and ecological divergence (Körner, 2004). In CR areas, geographical isolation and environmental gradients jointly promote genetic divergence among plant populations (e.g. Dantas-Queiroz *et al.*, 2021). Moreover, climate cooling from the late Oligocene to the middle Miocene expanded open-vegetation areas in the Neotropics (Jaramillo, 2023), allowing the establishment of plant populations into novel areas, ultimately facilitating plant speciation (Guillory *et al.*, 2024). In CR areas, endemic plant species have been inferred to expand geographically during the Last Glacial Maximum (Dantas-Queiroz *et al.*, 2023), suggesting that previous climate cooling might have impacted plant species. In this sense, mountain colonization and climate cooling have been inferred jointly to promote species diversification in other Neotropical plant lineages. For instance, the centropogonid clade of bellflowers (Campanulaceae) rapidly diversified after colonizing high-altitude areas in the Andes during the late Miocene (Lagomarsino *et al.*, 2016). Likewise, the Merianieae tribe (Melastomataceae) underwent faster species diversification after colonizing Andean regions during the middle Miocene (Dellinger *et al.*, 2024). Thus, we propose that ancestral *Chamaecrista* species experienced faster diversification when colonizing CR areas, probably owing to increased geographical isolation and environmental heterogeneity, concomitantly expanding across these montane areas during the climate cooling between the early Oligocene and the middle Miocene.

Topographic complexity enhances plant endemism in CR areas

Supporting our third prediction, CR-endemic *Chamaecrista* species have originated under greater geographical isolation than non-endemic species. This pattern indicates that CR areas have facilitated geographical speciation better than lowland areas, as expected by the MG hypothesis (Mosbrugger *et al.*, 2018; Muellner-Riehl, 2019). Given that extant CR-endemic and non-endemic *Chamaecrista* species originated primarily during the Plio-Pleistocene (Fig. 5A), their different speciation modes would be more likely to reflect their distinct geographical distributions, rather than time of origin. In this sense, CR areas could facilitate geographical speciation owing to their pronounced topographic complexity (Silveira *et al.*, 2016), arising from the underlying montane landscape (Saadi, 1995). CR areas are separated by valleys and mountain ridges, forming an 'archipelago-like' system that reduces the spatial connectivity among plant populations (Guillory *et al.*, 2024). Hence, multiple plant species display a low zygotic gene flow among their populations, indicating a limited seed dispersal among CR areas (e.g. Fiorini *et al.*, 2019; Dantas-Queiroz *et al.*, 2021). Consequently, population genetic divergence is associated with geographical distance for some CR plant species (e.g. Fiorini *et al.*, 2019; Dantas-Queiroz *et al.*, 2021). Nonetheless, other CR plant species also have their population genetic divergence associated with bottleneck effects in the Pleistocene (e.g. Collevatti *et al.*, 2012; Barres *et al.*, 2019; Silva *et al.*, 2020). Therefore, the pronounced geographical speciation among CR-endemic *Chamaecrista* species is likely to derives from the topographic complexity of CR areas, with some potential contribution from the Plio-Pleistocene climatic oscillations, as previously elaborated.

Beyond promoting geographical speciation, topographic complexity has probably hindered the dispersal of CR-endemic *Chamaecrista*. Topographic complexity has been assumed to hinder plant dispersal among CR areas greatly (Silveira *et al.*, 2016), which would promote great floristic differences among neighbouring CR areas (e.g. Echternacht *et al.*, 2011). Under this scenario, CR-endemic *Chamaecrista* species would be less likely to disperse to novel areas, consequently developing smaller geographical ranges than their non-endemic relatives with similar niche breadths (Fig. 3B). Topographic complexity is likely to impact the dispersal of other plant groups in CR areas, regardless of their dispersal mode. For instance, *Miconia* Ruiz & Pav. (Melastomataceae) and *Myrcia* DC. (Myrtaceae) produce berries dispersed by birds that presumably would enhance dispersal range, yet both plant groups have narrowly endemic species in CR areas (Santos *et al.*, 2017; Caddah *et al.*, 2022). Intriguingly, the geographical distance predicts a small proportion of the plant species turnover among CR areas (Barros-Souza *et al.*, 2026), suggesting a minor role of dispersal limitations in the assembly of CR plant communities. However, linear geographical distances become poor proxies for dispersal limitations in topographically complex landscapes where spatially close sites can have low connectivity owing to pronounced physical barriers, i.e. isolation by resistance (McRae, 2006). Hence, topographic complexity has probably imposed dispersal limitations and strongly limited the geographical range of CR-endemic *Chamaecrista* species, ultimately enhancing plant endemism in CR areas.

Geographical isolation in CR areas has also been associated with the prevalence of short-range dispersal modes

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(Silveira *et al.*, 2016), which would be advantageous in heterogenous landscapes (e.g. Hopper, 2009; Hopper *et al.*, 2021). Short-range dispersal becomes selectively advantageous when suitable habitats are spatially scattered over heterogeneous landscapes (Brown and Pavlovic, 1992). Under this scenario, short-range dispersal would prevent seeds from reaching unsuitable habitats and favour those remaining in suitable habitats, ultimately leading to habitat specialization (Brown and Pavlovic, 1992). Reflecting this theoretical scenario, CR plant communities have sparse and spatially constrained seed rains, which have been attributed to the prevalence of unassisted (autochoric) dispersal modes (e.g. Arruda *et al.*, 2021). Nonetheless, both CR-endemic and non-endemic *Chamaecrista* species have explosive pods, suggesting that autochoric dispersal modes could not account for the greater geographical isolation of this plant group in CR areas. Likewise, autochoric dispersal modes are present in CR-endemic and non-endemic species of multiple plant clades, such as *Barbacenia* Vand. (Velloziaceae), *Microlicia* D. Don. (Melastomataceae) and *Minaria* T.U.P. Konno (Apocynaceae) (e.g. Ribeiro *et al.*, 2014; Cabral *et al.*, 2021; Versiane *et al.*, 2021). Hence, short-range dispersal modes might pre-date the colonization of CR areas, subsequently becoming advantageous in the heterogeneous landscapes of CR areas (Silveira *et al.*, 2016). Thus, we propose that short-range dispersal modes might have facilitated the evolution of environmental specialization for CR-endemic *Chamaecrista* species, without necessarily enhancing geographical isolation among them.

The campos rupestres *as a montane domain*

The MG hypothesis explains plant endemism in mountain regions by: (1) greater specialization owing to the environmental heterogeneity of montane landscapes; (2) faster species diversification owing to 'species-pump' effects during climatic oscillations; and (3) greater geographical isolation owing to the topographic complexity of montane landscapes. As previously elaborated, we found that CR-endemic *Chamaecrista* species have evolved towards environmental specialization, also undergoing speciation under pronounced geographical isolation. We also found that CR-endemic *Chamaecrista* species underwent faster diversification upon colonizing CR areas between the early Oligocene and middle Miocene, rather than during the Plio-Pleistocene climatic oscillations. Hence, our results suggest environmental heterogeneity and topographic complexity as major promoters of plant endemism in CR areas, but they counter a 'species-pump' effect during recent climatic oscillations. Likewise, plant species richness is directly associated with elevation range in multiple mountain regions worldwide, but not with recent climatic variation (Muellner-Riehl *et al.*, 2019). Thus, we found partial support for the MG hypothesis in explaining the evolution of plant endemism in CR areas.

Nonetheless, we propose an alternative scenario for the evolution of CR-endemic *Chamaecrista* species. First, ancestral *Chamaecrista* species from lowland areas, probably a formerly expanded Amazon domain, dispersed to CR areas during the late Eocene (Rando *et al.*, 2026) or between the early Oligocene and middle Miocene (this study). These dispersals to CR areas might have occurred across distinct montane regions in Central, Eastern and Northeastern Brazil. When dispersing to CR areas, ancestral *Chamaecrista* species experienced faster diversification rates, probably owing to increased geographical and environmental divergences imposed by the montane landscape. This ancestral diversification burst gave rise to early CR *Chamaecrista* species able to occupy habitats inside and outside the montane landscape. Second, early CR *Chamaecrista* species faced selective pressures favouring environmental specialization, probably attributable to the environmental heterogeneity within CR areas. Such selective pressures have progressively constrained the potential for persistence and adaptation in lowland habitats, leading to the evolution of CR-endemic *Chamaecrista* species from the late Miocene onwards. Third, early CR *Chamaecrista* species also faced pronounced geographical isolation, probably attributable to the topographic complexity among CR areas. This topographic complexity has facilitated geographical speciation and constrained the geographical range of early CR *Chamaecrista* species, further impacting CR-endemic *Chamaecrista* species from the late Miocene onwards. Throughout, CR and CR-endemic *Chamaecrista* species progressively experienced slower diversification rates, reaching very low diversification rates during the Plio-Pleistocene. Therefore, an ancestral diversification burst upon colonization followed by long-term environmental specialization and geographical isolation have jointly promoted the evolution of endemic *Chamaecrista* species in CR areas.

These evolutionary processes have previously been suggested under other theoretical frameworks (e.g. Silveira *et al.*, 2016; Vasconcelos *et al.*, 2020). However, our study explicitly tested some evolutionary processes that were assumptions under these other frameworks. For instance, we inferred that CR-endemic *Chamaecrista* species originated under pronounced geographical isolation, supporting CR areas as an 'archipelago-like' system (Vasconcelos *et al.*, 2020). Likewise, we inferred that CR-endemic *Chamaecrista* species evolved towards environmental specialization under selective pressures, which have been assumed to drive plant species evolution in CR areas (Silveira *et al.*, 2016). Moreover, our study highlighted that CR-endemic species have undergone evolutionary processes that are linked with mountains, further suggesting that CR areas could be interpreted as a non-contiguous montane domain in Brazil. In this sense, the evolutionary dynamic of CR areas resembles those of other montane domains in the Neotropics, such as the Atlantic Forest and the Andes (e.g. Lagomarsino *et al.*, 2016; Nery *et al.*, 2023). Additional studies on other CR-endemic plant groups are necessary to verify the generality of this montane-like evolutionary dynamic in CR areas. Nonetheless, our study supports that CR areas constitute a distinct biogeographical unit (Colli-Silva *et al.*, 2019), whose evolutionary dynamic differs from the surrounding Cerrado and Caatinga (Simon *et al.*, 2009; Fernandes *et al.*, 2022).

SUPPLEMENTARY DATA

Supplementary data are available at *Annals of Botany* online and consist of the following. Figure S1: the MCC tree of *Chamaecrista* species by Rando *et al.* (2026). Figure S2: the

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interquartile range of hypervolume dimensions and their relationship with local Moran's *I*. Figure S3: net diversification rates for the MCC tree of *Chamaecrista* species, based on the best-fitting MiSSE model. Table S1: the seven evolutionary models fitted to the hypervolume size of *Chamaecrista* species. Table S2: *Chamaecrista* species and their respective geographical states, number of curated records (*n*) and performance metrics for hypervolume model.

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